Energy Policy 193 (2024) 114303

Contents lists available at ScienceDirect

Energy Policy

journal homepage: www.elsevier.com/locate/enpol

The temporalities and externalities of ancillary infrastructure in large-scale renewable energy projects: Insights from the rural periphery

Clemens Greiner [a,*], Britta Klagge [b]

[a] Global South Studies Center (GSSC), University of Cologne, Germany
[b] Department of Geography, University of Bonn, Germany

ARTICLE INFO

Keywords:
Infrastructure
Renewable energy
Planning
Co-benefits
Acceptance
Geothermal
Kenya

ABSTRACT

Ancillary infrastructures such as roads, workers camps, or water management systems are an essential prerequisite for realizing large-scale renewable energy projects. They usually are established before the power plant is constructed and differ from it in their temporalities and externalities. Especially in peripheral regions in the Global South, ancillary infrastructure can have negative impacts on the affected communities as well as bring them significant co-benefits. Despite its extensive and often far-reaching nature, as well as social and ecological consequences, ancillary infrastructure has received little scholarly attention, and is often not given sufficient consideration by planners and investors. This article foregrounds ancillary infrastructure in peripheral rural areas with little pre-existing infrastructure. It provides a typology of ancillary and other infrastructures as well as a systematic discussion of its temporalities and externalities, and argues that local acceptance can increase significantly owed to positive externalities deriving from ancillary infrastructure. We illustrate our conceptual framework with the example of a large geothermal project in Kenya. Based on these conceptual and empirical findings, we conclude with detailed policy recommendations.

1. Introduction

Large-scale renewable energy projects (LSREPs) are associated with various types of infrastructure that have different functions. We distinguish between primary, compensatory and ancillary infrastructures. In this contribution, we foreground ancillary infrastructure. This refers to new and/or retrofitted infrastructure of various types, such as access roads or water supply, necessary to implement and realize LSREPs, but not a direct part of the energy producing or transmitting complex. The socio-economic impact of ancillary infrastructure has received little attention in the scholarly literature (Ahmed and Gasparatos, 2020). This is surprising, since their importance tends to grow with the massive shift to more spatially extensive renewable energy sources, particularly in remote rural areas (Grubert and Zacarias, 2022; McCarthy, 2015; O'Sullivan et al., 2020), which are often characterized by a lack of appropriate road, water, electric, and/or other service infrastructures (van de Ven et al., 2021). The socio-economic and environmental impact of ancillary infrastructures tends to be significant in such regions (Gasparatos et al., 2021; Kramarz et al., 2021; Rudman et al., 2017). This contribution is thus a step toward closing the knowledge gap.

We argue that, in order to better understand the socioeconomic impacts of ancillary infrastructures, it is essential to sustain a focus on their temporalities (Ramakrishnan, O'Reilly, and Budds, 2020; Küpers and Batel, 2023). More concretely, we argue that their temporal dynamics may significantly deviate from the so-called U-curve of acceptance, which posits that local acceptance of LSREPs declines with implementation and increases after completion of the renewable energy project, namely, in the operation phase (Rudolph and Clausen, 2021; Wolsink, 2007; Wilson and Dyke, 2016). In contrast, we assume that ancillary infrastructure can be associated with benefits for the local population, which can play a positive role in acceptance during implementation, while poor planning, inadequate execution or failure to take into account the long-term consequences of ancillary infrastructure can jeopardise the acceptance of the overall project even after completion. We illustrate our point with empirical insights into a geothermal energy project in Kenya, which demonstrates the crucial role of ancillary infrastructure, their temporalities, and socio-economic impacts on affected communities.

In what follows, we first distinguish between the different types of infrastructure (primary, compensatory, ancillary) and explain our approach to the notion of temporality with reference to scholarly literature (2). This is followed by a typology that distinguishes between permanent and temporary ancillary infrastructure typically associated

* Corresponding author.
E-mail addresses: clemens.greiner@uni-koeln.de (C. Greiner), klagge@uni-bonn.de (B. Klagge).

https://doi.org/10.1016/j.enpol.2024.114303
Received 24 January 2024; Received in revised form 12 August 2024; Accepted 13 August 2024
Available online 18 August 2024
0301-4215/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Explanation of acronyms

CSR	Corporate Social Responsibility
EIA	Environmental Impact Assessment
ESIA	Environmental and Social Impact Assessment
GDC	Geothermal Development Corporation
KeNHA	Kenya National Highway Authority
KeRRA	Kenya Rural Roads Authority
LSREP	Large-Scale Renewable Energy Project
NEMA	National Environmental Management Authority
NGO	Non-Governmental Organisation
PV	Photovoltaics
SCORE	Sarawak Corridor of Renewable Energy
SIA	Social Impact Assessment

with LSREPs and a brief review of their temporalities as well as related positive and negative externalities (3). After introducing the context and methodology of our case study (4), we show how ancillary infrastructure plays a key role in understanding present and future livelihood changes in a geothermal project area in Kenya (5). In our discussion, we generalize from our findings (6) and, finally, explore the resulting policy implications (7).

2. Ancillary infrastructure and temporality: overview and definitions

The concept of ancillary infrastructure is marked by inconsistent terminology and definitions in the scholarly literature. To our knowledge, there is no widely agreed upon definition of ancillary infrastructure, nor have there been any systematic literature reviews.[1] While the term ancillary infrastructure is used in some publications (Kelly-Richards et al., 2017; Gasparatos et al., 2021), other authors use "establishing infrastructure" (Bakken et al., 2012), "complementary infrastructure" (Schulz and Saklani, 2021), "itinerant infrastructure" (Kelly, 2019) or "support infrastructure" (Tang et al., 2013). The lack of a clear terminology presents a challenge in reviewing existing research; our review (3) thus largely reflects isolated mentions and scattered evidence.[2] Furthermore, local socio-economic impacts of ancillary infrastructure also tend to differ depending on the energy source. Wind farms, for example, tend to have a significant part of their economic impact during the construction phase, while biomass needs constant input during operation (del Río and Burguillo, 2008).

The definition and review of ancillary infrastructure is further complicated by the fact that scholarly research into the socio-economic impacts of large-scale renewable energy projects is scarce (Gasparatos et al., 2021; Terrapon-Pfaff et al., 2019), although there is a growing body on "critical renewability" (Knuth et al., 2022) and local acceptance (Rodríguez-Segura et al., 2023). A deeper engagement with these bodies of literature is beyond the focus of this article. Suffice here to state that these and other conceptual frameworks for either assessing the impact of renewable energy deployment on local sustainability (del Río and Burguillo, 2008), or for systematizing temporal dimensions in social acceptance research (Küpers and Batel, 2023), do not differentiate between ancillary and other infrastructures. Furthermore, ancillary infrastructure and compensatory infrastructure is sometimes lumped together under the notion of 'in-kind benefits' and therefore not clearly distinguishable. To begin, we therefore explain our threefold distinction of LSREP-related infrastructure into primary, compensatory, and ancillary infrastructure.

Primary infrastructure directly contributes to power generation and transmission. It includes power-generation facilities as well as the power lines required to connect power generation with the national grid. While the latter is the same for all renewable-energy sources, the former can differ depending on the renewable energy source and its operational characteristics. While solar PV parks and wind farms are rather self-contained, hydropower and geothermal energy projects consist of both conventional turbines and additional facilities, like large-scale dams or geothermal steam pipeline networks. Such facilities have a much larger spatial and often also ecological footprint than wind or solar PV facilities (Pasqualetti, 2011).

Compensatory infrastructures include, first of all, facilities that are established as part of resettlement schemes, especially housing, schools, health infrastructures and access roads to the new living sites of resettled people. But even without resettlement schemes, LSREPs often include compensatory infrastructures to improve local living conditions and thereby achieve acceptance of, and mitigate conflict with, affected local populations. Examples include the establishment or upgrading of educational, health, and cultural facilities. Such compensatory infrastructure is usually paid for by the investor, sometimes with support from local or national state agencies, or international donors. In many countries, compensatory infrastructure is required by legal processes, such as environmental and social impact assessments, but it may also be voluntary and part of the investor's corporate social responsibility (CSR) policies.

Ancillary infrastructure encompasses all facilities that are technically necessary to realize a LSREP. In contrast to other authors (Tang et al., 2013; Kelly-Richards et al., 2017), we do not include the electric grid and its components. Ancillary infrastructure can take different forms. Most importantly, all types of LSREPs require access via roads; construction activities are especially dependent on wide roads that allow for the transportation of heavy equipment in large trucks. Ancillary infrastructure also encompasses various facilities that are directly related to construction activities, including temporary storage facilities (e.g., for building material and fuels), repair facilities and workshops, as well as housing and catering facilities for workers, managers, and engineers. Furthermore, some LSREPs, and geothermal in particular, are also dependent on the large-scale provision of water, particularly for drilling in the exploration and construction phase. The more peripheral the location of a LSREP, the more ancillary infrastructure needs to be built, and the greater its potential to significantly impact local livelihoods and ecologies (Gasparatos et al., 2021).

The importance of ancillary infrastructure varies with project phases. In what follows, we refer to the different phases of LSREPs, starting from planning and exploration, through construction and implementation, to operation and finally decommissioning. Ancillary infrastructure is most important in the exploratory and construction phase, whereas once exploration and construction are finished and operation has started, most ancillary infrastructures are no longer required, or are only required on a much smaller scale for operation and maintenance. However, different types of ancillary infrastructure, especially roads versus other kinds of ancillary infrastructure directly related to construction, like worker camps and storage facilities, might differ in their temporalities, as we explain in the next section.

Before we trace some of the evidence for temporalities and

[1] We would like to highlight that our focus in this contribution differs from life-cycle assessments of LSREPs (Gasparatos et al., 2021), since we are not concerned with the mining and sourcing of materials, the production of components, or the costs of decommissioning and disposing of LSREPs. We focus primarily on implementation and, to some extent, on the generation phase, taking infrastructural ruination into account (Howe et al., 2015).

[2] Our literature review is based on Google scholar searches with keywords "renewable energy", "ancillary infrastructure", "support infrastructure", "implementation infrastructure", "community benefits", "access roads", "co-benefits", among others, as well as snowball sampling of other literature from the more promising articles. Due to the sparseness of scholarly studies, we do not distinguish between energy sources, nor do we differentiate between studies focussing on the Global North (relatively many) and the Global South (relatively few).

externalities of ancillary infrastructure in the scholarly literature (3), let us briefly explain our concept of temporality. By temporalities we refer to the various timeframes that govern the envisioning, exploration, planning, construction and implementation, operation and eventually decommissioning of these infrastructures, including their ruination, maintenance, and retrofitting (Howe et al., 2015; Ramakrishnan, O'Reilly, and Budds, 2020; Brookes et al., 2017).

Recent years have seen a boom in the literature on infrastructural planning and time (Abram, 2011, 2014; Abram and Weszkalnys, 2011, 2013; Bear, 2016; Hetherington, 2017; Howe et al., 2015). In anthropological approaches, planning is characterized as "an inherently optimistic and future-oriented activity" (Abram and Weszkalnys, 2011: 3; Laurian and Inch, 2018), an activity that deals with the temporal transition from current states to desired states. Almost by default, this results in discrepancies between different expectations and their time frames, which can be described as clashes between "technocratic and lived time" (Abram and Weszkalnys, 2011: 14). This is a crucial point, since once infrastructure projects are in the planning phase, they are often loaded with promises and expectations for a better future (Larkin, 2013). Research on megaprojects (Flyvbjerg, 2014; Ansar et al., 2016), however, demonstrates that long-term benefits are usually grossly overestimated, while costs tend to escalate: "when reality catches up with optimistic, or manipulated, estimates of schedule, costs, or benefits; delays, cost overruns, and benefit shortfalls follow" (Flyvbjerg, 2014: 12).

We therefore argue that the different temporalities, promises, and expectations associated with LSREPs are important to better understand the externalities of ancillary infrastructures, as well as their potential for giving rise to conflicts between investors, governments, and local populations. More concretely, we argue that in order to achieve a comprehensive assessment of LSREPs and their impacts on local livelihoods, it is necessary to develop a better understanding of ancillary infrastructure and its potential for socio-economic and socio-ecological change and conflict. Only then can the future development of LSREPs meet the demands of procedural sustainability approaches (del Río and Burguillo, 2008).

3. Review: temporalities and externalities of (ancillary) infrastructure: reversing the U-curve?

Although research on the temporality of LSREPs has gained importance (Labussière and Nadaï, 2018; Delina and Sovacool, 2018), it remains a largely underexplored field (Rudolph and Clausen, 2021; Küpers and Batel, 2023). Among the earliest approaches on time and renewable energy projects is the 'U-curve' concept (Rudolph and Clausen, 2021; Küpers and Batel, 2023; Wolsink, 2007). The convex curve suggests that local acceptance for renewable energies is high before a project is announced in a given implementation area, falls drastically and is lowest during its implementation, and tends to increase sometime after a project is finally implemented. The concept rests on many unexplored assumptions, particularly regarding long-term post-implementation acceptance. Rudolph and Clausen (2021) thus refer to it as "the elusive U-curve". Furthermore, the concept lacks explanatory potential and fails to shed light on the complexities of people's responses (Küpers and Batel, 2023).

We argue that closer attention to ancillary infrastructure and its temporalities can reveal interesting contradictions to this model, at least for LSREPs in remote rural areas. In short: local acceptance can increase significantly during implementation owed to positive externalities deriving from ancillary infrastructure, such as road upgrades. However, it can also drop once negative externalities of ancillary infrastructure become manifest, and ancillary infrastructure is no longer maintained because of its temporary character. To further explore this relation, we distinguish between two types of ancillary infrastructure (Table 1). First, we refer to infrastructure that is meant to be permanent, such as roads, water, or telecommunication. This infrastructure provides the basis for the development of the project area. This type of infrastructure is potentially long-lasting, because it is not dismantled after construction is finished. However, if not maintained, such as by investors or state agencies, it is in danger of decay and may lose functionality after some time. Second, and by contrast, temporary ancillary infrastructure is built for a limited time, usually only for use during the construction phase, and is thus removed after the project is completed, thereby contributing and leading to a 'boom and bust' activity pattern and temporality.

As for the different externalities of LSREPs' ancillary infrastructures, we provisionally classify them into positive and negative, while acknowledging that such normative judgements are highly dependent on positionality. Both types of ancillary infrastructure, permanent and temporary, may lead to local, short-term economic boosts and temporary job creation, often for unskilled labour, especially during the construction phase of a project (Guerin, 2019; Callaghan and Williams, 2014; Lema et al., 2021). Such effects may be more severe than the impact of the project itself (Laurance et al., 2015).

In addition to construction-related short-term effects, permanent ancillary infrastructure, such as roads or telecommunication facilities, also enhance mobility and access to previously remote areas. This may benefit communities, since it improves access to services and markets in areas that have previously been difficult to access (Crootof et al., 2021; Góes et al., 2021; Brunet et al., 2020). However, roads in particular may also be associated with negative external effects, such as the loss of biodiversity, increased accident rates, or general disturbance by traffic (Rudman et al., 2017; Kelly, 2019), alongside the influx of unwanted actors, such as land speculators or the increasing presence of repressive governments in previously isolated areas (Aung et al., 2021). Furthermore, as Sovacool and Bulan (2012) report from the Sarawak Corridor of Renewable Energy (SCORE) in Malaysia, access roads may open forests to illegal logging and hunting; in the case of hydropower in Norway, other scholars report that roads bring severe disturbances for Sámi reindeer herders (Engen et al., 2023).

Temporary ancillary infrastructure, particularly workers camps, may open up new income-earning opportunities for local populations, particularly in the service sector. However, this often goes along with the risk of negative externalities such as drug abuse, prostitution, sexual harassment, violence, and conflict. Such unwanted externalities are reported, for example, from the above-mentioned SCORE project (Sovacool and Bulan, 2012), from the Lake Turkana Wind Power Project in Kenya's arid north (Danwatch), from the Estreito hydroelectric plant in Brazil (Catolico et al., 2021), and from a range of Canadian hydroelectric projects (da Silva et al., 2021).

In light of such challenges and dangers, the range of potential

Table 1
Temporalities and externalities of permanent and temporary ancillary infrastructure.

	Permanent ancillary infrastructure; e.g., roads, water, telecommunication	Temporary ancillary infrastructure; e.g., worker camps, storage facilities, workshops
Temporality	*Potentially long-lasting*, but danger of decay	Mostly temporary, construction-related *'boom & bust' activity pattern*
Positive externalities	(Temporary) job creation for unskilled labour, income generating opportunities in services (e.g., bars/restaurants, shops) Enhanced access to services, markets, and facilities	
Negative externalities	Loss of land and biodiversity, easier access for locally unwanted actors such as police and military forces or land speculators	Increased demand for locally problematic goods & services such as drugs and sex services, social conflict

impacts of ancillary infrastructure appears to have been neglected in the existing literature on local socio-economic impacts of LSREPs. Similarly, such impacts also seem to be underreported in environmental and social impact assessments (ESIAs), as Kelly-Richards et al. (2017: 257) observe in regard to externalities of construction and ancillary infrastructure in the case of hydroelectric schemes (see also Bakken et al., 2012; Corry, 2011). This is all the more problematic since, at least with respect to road construction in tropical countries, environmental impact assessments (EIAs) generally tend to suffer from structural deficiencies and are therefore not well suited to cover wider aspects of many road projects (Alamgir et al., 2017).

With the following case study on geothermal energy developments in Kenya, we will provide a detailed elaboration of potential – positive and negative – externalities of ancillary infrastructure to illustrate our arguments.

4. Context and methodology

The Baringo-Silali Geothermal Project is located in the Baringo basin of Kenya's Central Rift Valley. It is a Government of Kenya project, sponsored by the German development bank KfW. Exploration studies started in 2010 and estimated a potential of up to 3.000 MW available from three extinct volcanoes on the Rift Valley floor, Paka, Korosi and Silali. The ecology of the area is characterized by semi-arid acacia bush savanna, shaped by high inter-annual variations in rainfall and recurrent droughts. The area is classified as Arid- and Semi-Arid Lands (ASAL). Lake Baringo, one of the Rift Valley's freshwater lakes, is the only nearby perennial source of water. Administratively, the project area belongs to Tiaty East Subcounty, which is part of Baringo County. Tiaty East is sparsely settled by Nilotic-speaking Pokot, who have been living from semi-nomadic pastoralism for much of the past 200 years (Vehrs, 2022). Since about the turn of the millennium, however, an increasing number of households have settled more permanently and engage in rain-fed cultivation in addition to livestock husbandry (Greiner et al., 2021b). The area was marginalized since colonial times with high poverty and illiteracy levels and strong population growth rates. Frequent outbreaks of violence and cattle raids between Pokot people and their neighbours, particularly Turkana, Ilchamus, Tugen, and Samburu, have worsened the situation (Greiner, 2013).

The empirical study we present is grounded in long-term and ongoing ethnographic fieldwork in the above-mentioned area (since 2009). This includes qualitative interviews, ethnographic observations and other methods, which provide deep insights into local culture, livelihoods and conflict dynamics. The long-term nature of ethnographic field research makes it possible to build trusting relationships with local interlocutors and to track changes in socio-cultural (and in this case also socio-ecological) patterns over time (Bollig et al., 2014; Greiner et al., 2021b). Since 2017, this work is complemented by perspectives from economic geography through a joint research project on "Energy Futures", as part of a larger research consortium on future-making and socio-ecological transformation in rural Africa.[3] "Energy Futures" explores renewable energy development in Eastern and Southern Africa, with a particular focus on geothermal energy production in Kenya. In Kenya, joint visits were repeatedly made to key sites of geothermal development (Olkaria, Menengai, Suswa, and Baringo-Silali) as well as in Nairobi, Naivasha and Nakuru, where experts, practitioners, politicians, and financiers in the geothermal sector work. Based on our site visits, networking through participating in conferences and other events as well as snow-balling over a period of several years, we identified suitable interview partners and conducted interviews with key government experts, especially at the county level (I_GovCty), in state agencies, most notably the Geothermal Development Corporation (GDC) (I_GDC) and in development finance institutions (I_DFI). In addition, we interviewed members of local communities affected by geothermal development (I_Comm). This was complemented by document analysis and in-depth research by PhD and MA students engaged in the project. The long-term nature of our research has enabled us not only to build trust-based relationships with many of the people we have repeatedly interviewed. It has also allowed us to repeatedly cross-check and triangulate our data, generating a track record of publications on different aspects of geothermal development in Kenya (see, e.g., Greiner et al., 2021a; Greiner et al., 2023; Klagge et al., 2020).

[3] Future Rural Africa. Future-making and social-ecological transformation (https://www.crc228.de).

5. *Ancillary infrastructure for geothermal development: The case of Baringo-Silali, Kenya*

The exploration and development of geothermal resources in Baringo-Silali in Tiaty East was only possible with the large-scale development of ancillary infrastructure. Until the arrival of the geothermal project, the region was poorly developed in terms of roads and water infrastructure. The development of the Baringo-Silali Project, including establishing necessary infrastructure, was entrusted to the Geothermal Development Corporation of Kenya (GDC), a parastatal organisation designed to de-risk and promote geothermal resource exploration. The GDC is responsible for the development and maintenance of steamfields, including from the point that steam is sold to a power producer (for more on this, see Klagge and Nweke-Eze, 2020). Following successful surface explorations, the GDC started to develop a substantial road and water network from 2015, in part through subcontracting companies. Ancillary infrastructure was therefore planned and built at a time when the success of the project was highly fragile and financially risky (Klagge, 2021), as the potential predicted by the surface studies had to be first confirmed by exploratory drillings. For these drillings, however, access roads, reliable water supplies, as well as other temporary ancillary infrastructures were needed, among other things. Employing the distinction between permanent and temporary ancillary infrastructure (Table 1), we now present in detail the empirical insights we gained from our case study on the Baringo-Silali geothermal project by first outlining the situation of, and changes to, roads (5.1) and water provision (5.2) as well as their impacts on local livelihoods. This is followed by a brief section on temporary ancillary infrastructure and local employment in the context of the project (5.3). Based on our empirical findings we then discuss externalities and temporalities of ancillary infrastructures in section 6.

5.1. *Roads as ancillary infrastructure in Baringo*

It takes about 100 truckloads to transport a drilling rig, not including the transportation of other temporary infrastructure, such as camp facilities for workers and engineers. Roads are thus crucial for the project's success (I_GDC). The only main road to Tiaty East is the north-south road (B4), which connects to the city of Nakuru and ultimately to Nairobi and Mombasa, and was only paved in the early 1980s (Greiner et al., 2021a). Until 2015, however, the asphalt was in very bad shape, bridges were defunct or missing, and, north of the centre of Loruk, the tarmac gave way to gravel. Road construction and maintenance is a highly politicized affair in Kenya (Kochore, 2016). None of our interviewees could (or would) tell us what strings were pulled to release the budget that enabled the Kenya National Highway Authority (KeNHA) to carry out the much-needed upgrade and extension of the B4 for GDC activities in Baringo. However, conversations with senior GDC engineers and Baringo county officials suggested that there was a clear link to the national plan for geothermal development (I_GDC; I_GovCty).

Whereas KenNHA is responsible for national roads (i.e. A- and B-roads), regional and feeder roads in rural areas fall under the mandate of either the Kenya Rural Roads Authority (KeRRA) or the county. In the case of the Baringo-Silali geothermal project, however, the GDC was responsible for building and upgrading access roads to the project

regions (I_GDC; I_GovCty). Since the project started, approximately 155 km of such roads have been cut through an area that was basically roadless before. The new roads usually follow former foot- and livestock paths, and their routes are discussed and agreed upon through committees formed by local elders, as GDC engineers and community relations personnel emphasize (I_GDC; I_Comm). The Baringo county government, however, has no legal mandate to conduct a prior environmental assessment, which would ensure, for example, that the new roads do not impact negatively on wildlife, water catchments, or forested areas. As one interviewee explained: "They [the GDC] identify the roads, they ensure local participation and their own environmental assessment (…) they are regulated by the national government" (I_GovCty). The KfW and the GDC have given contradictory answers to the question of whether and, if so, what form of impact assessment was carried out before the road and other infrastructure was expanded (I_DFI; I_GDC). Since at least one of the new roads (leading to Silali) connects to Turkana County, it will be classified as B-road and fall under KeNHA responsibility, which requires ESIAs for its projects (I_GovCty).[4] Apparently, there was an assessment for the National Environmental Management Authority (NEMA), but despite repeated requests, we have never secured access to any official documents. Unlike the comprehensive ESIA, which is mandatory before the construction of the power plant, the construction of extensive ancillary infrastructure apparently does not require such study.

Ethnographic observations show, that the externalities of the new roads in Baringo-Silali, are substantial, in both positive and negative senses, and they have changed the social and physical environments from the very beginning. They enable the easier influx of people, goods, and ideas, especially via the ubiquitous *Pikipiki*, or motorbike taxis. Thus, first of all, the roads contribute to more interregional communication and experiences, to new aspirations, and changing attitudes (Greiner et al., 2021a). Second, they allow easier transport to health and educational facilities as well as to (wage) labour opportunities, all of which can have profound socio-economic impacts on local livelihoods (I_Comm; I_GDC). Third, roads attract other infrastructures, notably telecommunication boosters, but also state infrastructure, such as dispensaries and schools, as well as shops, bars, and restaurants, thereby providing new local sources of income (I_Comm). Fourth, roads increase the value of land and lead to land grabbing and elite capture (Greiner, 2020). Fifth, roads enable extraction of (precious) trees species, de-forestation, and charcoal production. "The first victims of the road are the trees", as one local observer tells us (I_Comm). Sixth, roads contribute to the fragmentation of the region and to an increase in accidents, particularly with livestock. Seventh, and finally, the roads are feared above all by the traditional parts of the population, as they improve access for the military and police (I_comm). This fear of the 'securitisation' of the area is plausible in view of the military operations against alleged cattle rustling in the region (Vehrs, 2022).

Initially, the new roads were not very welcomed by many residents of the area. "People fled the road, and only came back later, when they realized some of the benefits", as one GDC community relations officer put it (I_GDC). These benefits, as described above, are in enhanced transport, but most importantly in the new commercial centres that sprung up along the roads as well as the water infrastructure that follows the road (see below). For the time being, the responsibility for the maintenance of the new road network is secured. Since the GDC will continue to drill in order to ensure future steam supply to power plants, they are likely to be present in the area for the coming decades (I_GDC; I_GovCty). It is also their responsibility to maintain the road infrastructure, although they admit, "sometimes we lack equipment and manpower to do so" (I_GDC). These deficits are apparent, as some of the roads are in very bad shape, especially after the rainy season. "Ideally, maintenance is transferred to the county government or to KeRRA", as one GDC engineer tells us (I_GDC). However, the county government is not in a financial position to provide necessary road maintenance in the near future, and therefore they welcome the GDC's activities: "It really takes pressure from us. We consider the GDC activities in our annual planning, and sometimes we even ask them to help us with some roads", as one high-ranking county official explains (I_GovCty). However, it is unclear what will happen to the permanent ancillary infrastructure in areas that have been explored, but determined to lack steam potential and are thus abandoned by the GDC (I_GDC; I_GovCty).

5.2. Water situation and infrastructure in the project area

Tiaty is a dryland, and water is only accessible during and after the rainy season in some pan dams, through some boreholes (many of them saline), and from Lake Baringo. This was also a challenge for the geothermal project, since the drilling operations require a constant flow of water, and efficient water infrastructure is an indispensable prerequisite. Thus, after the roads were constructed, and initial exploratory drilling was scheduled to start in Paka in late 2018, all efforts were concentrated on finalizing the water infrastructure (I_GDC). To supply water to the area, water is pumped under high pressure and via booster stations from Lake Baringo (about 30 km away) to huge basins on top of Paka mountain, and – with support of an additional booster station – on Mount Silali, from which it is released by gravity for drilling. The GDC had originally planned to provide water trucks to the surrounding communities, which was also part of an early agreement (I_GDC). However, this proved to be unsatisfactory and led to complaints and roadblocks by locals, threatening the future of the project. Thus, the GDC finally changed course and decided to build 20 waterpoints to supply filtered water for nearby communities, and an additional 40 waterpoints for livestock, using the water infrastructure established for drilling (I_GDC; I_Comm). These were financed by the KfW and declared a corporate social responsibility project (Greven, 2023).

Fetching water for the household is viewed as part of women's work in Pokot society, and women often have to walk for miles to access the precious resource. Ethnographic insights show that the dynamics of this gendered labour have changed substantially in areas now served by the new community water points. Moreover, some of the livestock no longer needs to migrate to distant water sources; instead, there are now crowds at the new water points, which have also developed into meeting places. However, members of affected communities have also reported some concerns, notably fear that the new water points could lead to "the influx of more herders and livestock, leading to potential conflicts and diseases, degradation of pasture, and dependency on the GDC" (Greven, 2023: 242). Beside the fact that local people continue consuming unfiltered water from livestock watering points and from leaks in the pipelines, there is also the danger of pipelines exploding under high water pressure. Meanwhile, the warnings inscribed on pipelines ("Danger – Hatari. High pressure pipe") are not particularly helpful, given the high levels of illiteracy in the area. Nevertheless, some local people have established irrigated fields close to water points and the potential of future irrigation projects in this drought-prone area appears to be a key interest of the county government, which hopes that reduced reliance on livestock will reduce conflicts in the area (I_GovCty).

When GDC plans to build water infrastructure in the project area became known, this created high expectations among the local population (Greven, 2023; Greiner, 2020). From the outset, water was viewed as a way to win 'hearts and minds' for the project (I_GDC). However, the GDC's planning horizons are diverging from the high expectations of local communities. Under pressure to start operating, the GDC concentrated on building infrastructure that primarily suited their needs, leaving some communities behind in those areas that were scheduled for later drilling. Disappointments over failing water supplies and dry pipelines sometimes give way to unrest and accusations against the GDC (Greven, 2023), while the GDC mostly blames contractors for not

[4] https://kenha.co.ke/wp-content/uploads/2021/09/Strategic-Plan-for-2018-2019-to-2022-2023.pdf.

delivering diesel to water pumps (I_GDC). However, some of the water infrastructure appears makeshift, and thus it does not come as a surprise that some Pokot elders blame poor maintenance and improper construction work for the lack of water supply (Greven, 2023). Meanwhile, local communities are not simply waiting for things to change. They have adapted, manipulated, and even sabotaged parts of the water infrastructure for their own needs. These latter actions have led to higher maintenance costs, and motivated the GDC to establish local waterpoint committees in order to give the communities more ownership and control of the resource (Greven, 2023). New meeting places, small commercial centres, and irrigated gardens have emerged near water points and close to points of frequent leakages along the main pipelines. Discussions with the county government on future cooperation, including the possible takeover or expansion of infrastructures, have only just begun (I_GovCty; I_GDC). According to one high county official in charge of water, there are ongoing discussions about whether the county could build its own infrastructure to pump lake water, but, as the time of writing, they are planning to tap into the infrastructure provided by the GDC (I_GovCty).

5.3. Temporary ancillary infrastructure and local employment

In addition to a new road and water infrastructure, there is also temporary infrastructure involved in geothermal exploration and drilling. Beside storage facilities needed to store gear and materials, there are, most notably, the camps for rig workers and engineers. The camp for one rig alone hosts some 60 people working in shift around the clock and in shifts that change every fortnight (I_GDC). These camps have catering units, but, since the GDC is a parastatal organisation, they are not allowed to procure meat locally, but need to source it from the Kenya Meat Board to prevent corruption (I_GDC). Nevertheless, there are also local people who sell food and other goods near these camps. Although there is not much entertainment for workers who are not on shift, conflicts between workers and local community members have so far been rare. This may be due to sensitizing training offered by GDC community-relations workers, as a GDC community relations officer suggests (I_GDC). Measures also include HIV-AIDS sensitization and the free distribution of condoms. The camps also have a medical unit for workers, which is open to local populations, and their ambulance also transports local people to hospitals (I_GDC; I_Comm). Local work opportunities in the construction of road and particularly water infrastructure have fluctuated over time, and were mainly provided by contractors. Delayed payments for these jobs have been a major and persistent source of discontent during the project (I_Comm). While the number of casual construction jobs has decreased sharply after the ancillary infrastructure construction work was finished, the number of local people employed as guards for security services has steadily increased over time, due to a growing number of drilling sites (I_GDC). As a result, there is an increase in local income, which led to more local retail businesses, as we could observe on repeated visits over several years.

6. Discussion: externalities and temporalities of ancillary infrastructure

Our case study shows that ancillary infrastructures can have substantial positive, but also negative effects (externalities) for local populations, especially in the construction phase (temporality). Given the prior situation of roads, water, and employment in northern Baringo, the arrival of extensive ancillary infrastructure in the project area raised hopes but also concerns. The high hopes placed in the water infrastructure in particular had the potential to bend the 'U-curve' in favour of the project at the start of the construction phase, which is exactly what happened. Ongoing disappointment with the actual functioning of this crucial part of the ancillary infrastructure, however, may turn the tide against the project, even before the primary energy infrastructure starts operating. In this section, we discuss the importance of pre-construction analysis of both positive and negative externalities (6.1), then address challenges associated with the temporalities of permanent ancillary infrastructure (6.2), and with wage labour and temporary infrastructures (6.3), before concluding our contribution with a few policy recommendations (7).

6.1. Externalities of ancillary infrastructure: pre-construction analysis and the question of impact assessments

The various externalities of ancillary infrastructures in the Baringo-Silali geothermal project described above – both positive and negative – show that roads and water infrastructures especially can have profound socioeconomic impacts that can outlast exploration and construction phases. Since the concrete assessment is situated and contingent on local values and specific historical experiences, ancillary infrastructures should be centrally and carefully included in pre-construction analyses of LSREPs' potential impacts on local livelihoods. Such analyses should require engagement with local communities before and while an LSREP is established. In the Baringo-Silali project, for example, the potential for new water supplies to serve as an additional benefit for local communities was long underestimated and initially only developed hesitantly. This aligns with an observation from wind power projects in water-scarce northeastern Brazil, where Góes et al. (2021: 15359) observe that "the co-benefit of water management could be greatly improved."

Given the potential impact, it is surprising that comprehensive EIAs and ESIAs for ancillary infrastructure development appear to be lacking not only in the case of Baringo (see 3). This is not to say that ESIAs are the only appropriate approach for assessing long-term impacts. By contrast, reliance on such tools can even increase conflict, such as those observed in the case of Social Impact Assessments (SIA) in hydroelectric schemes in Canada (da Silva et al., 2021). SIAs, these authors conclude, are inadequate tools to prevent conflicts and to understand a given project's impact on culture, wellbeing, and livelihoods of affected communities, since they tend to overlook critical parameters pertaining to culture and livelihoods, particularly of indigenous communities. In the light of such challenges, and the potential for both damages and benefits, the inclusion of local elders into road planning, and the establishment of local water committees in the case of Baringo-Silali, are thus first steps to inclusive planning. However, while engaging elders (only) might be politically opportune – and indeed is a common historical pattern in many (post)colonial contexts – limiting engagement to (disproportionately male) elders may overlook important social dynamics, opportunities, and stakeholders, as women are disproportionately responsible for most household tasks, and young people are often better educated and connected to wider society. Therefore, broadly empowering, inclusive, and participatory approaches that include a diversity of members from the affected communities should be pursued (Becker et al., 2003).

6.2. Temporalities I: infrastructural planning, permanent ancillary infrastructure maintenance, and local adaptation

Ancillary infrastructures are planned and constructed when project success is most fragile and financially risky. During this time, they are given high priority, as only their smooth functioning enables the actual investment to take place according to plan. Some of these infrastructures are built relatively quickly and from scratch, with rather short time horizons (e.g., temporary storage sites or accommodations for workers). Yet, as is also the case with permanent ancillary infrastructure in Baringo (i.e. roads and water), maintenance issues prevail. This is an issue evident in other cases, as well. Access roads, for example, often appear to be built in a rough and dirty manner with no backup plans for long-term maintenance. This can lead to frustrations among local users who may initially see such roads as a co-benefit (Brunet et al., 2020; Boyer, 2019).

In the case of a hydroelectric dam in southern Chile, the poor construction of access roads even triggered landslides and erosion (Kelly, 2019), and residents in the vicinity of wind power developments in rural Spain and France complain that poorly maintained access roads may even contribute the spread of wildfires (Dunlap, 2021).

In addition to taking care of long-term maintenance, modification of ancillary infrastructure by local populations and/or adaptation to their needs must also be considered in the planning processes through more comprehensive community engagement and future-oriented planning. While the scholarly literature acknowledges that infrastructure is modified and reworked by its users (Lawhon et al., 2023), this dynamic needs to be included in designing and managing such infrastructures. In the case of Baringo-Silali, it appears to be the water system that is most affected by modifications, thereby providing extra-benefits for local users, but at the same time challenging the functionality of the overall system. The establishment of water committees to address this issue is a first step, but there could be additional co-benefits if local ideas, for example for irrigation projects, are incorporated into the system design of ancillary infrastructure from the outset.

The extent to which investors and state agencies may feel responsible for the long-term financing, operation, and maintenance of both ancillary and compensatory infrastructures can vary. Such commitments depend on underlying obligations, contracts, and promises, on the extent to which investors are integrated into the respective governance structures (as 'corporate citizens'), and on how much pressure the local populations (or international donors) can exert on investors and the state. Generally, the inclusion of additional stakeholders can help to avoid worst-case scenarios in which physical infrastructures fall into disrepair, and educational and health programmes are disrupted, thus leaving local populations with more problems than benefits. Sometimes, the costs of ancillary infrastructure, like the upgrading of an existing road network or its maintenance, might be shared with or taken over by government institutions, while donors could take responsibility for educational and health programs. Such sharing of responsibilities and financial support might change over time and should be part of more inclusive and future-oriented planning, thereby meeting the demands of procedural sustainability approaches (del Río and Burguillo, 2008)

6.3. *Temporalities II: wage labour and temporary ancillary infrastructure*

Wage labour is one of the major benefits for local populations from any larger infrastructure project. However, especially in LSREPs, once power generation starts, the demand for – mostly unskilled – local labour, which is usually high during the construction of ancillary, compensatory, and primary energy infrastructure, implodes. Overall, with construction finished and operation starting, there is the danger that benefits in terms of wages and other income-earning opportunities derived from new or improved infrastructures, such as for service provision (e.g., catering), will vanish or dwindle over time due to disruptions in financing or lack of maintenance, leaving local populations disappointed and with feelings of deception.

The problem of the temporary and unskilled nature of jobs for local populations is reported from many projects, for example, from hydroelectric projects in Nepal (Crootof et al., 2021), or from a large-scale solar power plant in Madagascar (Brunet et al., 2020). This is in line with the general observation that renewable energy is frontloaded and temporally uneven, with larger numbers in manufacture, development, and construction and – in the case of wind and solar – limited numbers in maintenance and operations (Pearse and Bryant, 2022). Against this background, it is all the more important to communicate clearly from the beginning, and make other (in-kind) benefits last as long as possible. Furthermore, un- and low-skilled jobs that remain after construction should be given to local people, possibly after on-the-job and other training in the construction phase to upskill the local labour force. Moreover, workers and their families could be supported in using earned income to invest in more long-term income-earning opportunities.

7. Conclusion and policy implications

While primary (energy) infrastructures are planned as long-term investments and maintained accordingly, this is not necessarily the case for ancillary infrastructures. Nonetheless, these infrastructures are of special importance during exploration and construction phases, as they are required to get energy infrastructure started. This is particularly the case in peripheral rural areas with little pre-existing infrastructure. While in many areas, benefits from primary infrastructure are not tangible for local people (cf. examples from wind infrastructure, e.g., Boyer, 2019), carefully planned ancillary infrastructure can help to increase acceptance of LSREPs and benefit local populations. Thereby, ancillary infrastructure can bend the 'U-curve' (Wolsink, 2007) – which posits a drastical fall of acceptance of local LSREP-projects during implementation – in favour of the project. This is, as our case stuy shows, especially the case for roads and water infrastructures, which not only provide labour opportunities, but also various co-benefits. However, neglect or poor planning of ancillary infrastructure can also lead to frustration being directed against the energy project itself.

However, beside labour opportunities, in-kind, or other co-benefits, ancillary infrastructure can also bring severe disadvantages for locally affected communities, adding up to the burden/footprint of primary infrastructure. Our analysis shows that the further development of positive externalities is still open in the case of geothermal in Baringo and dependent on how GDC and public actors will take care of maintaining the established infrastructures (Greiner et al., 2023). While too much emphasis on cost recovery may hinder the sustainable development of ancillary infrastructures, there are also ways that they can be made (more) beneficial for local communities, and thus sustainable in a long(er)-term perspective.

8. Policy recommendations

- Project developers and responsible authorities should take ancillary infrastructures into account from the beginning and integrate them into long-term project planning;
- Public authorities and project planners should consider long-term maintenance and secondary-use concepts well in advance, in order to ensure that sustainable and future-oriented financing and management structures will be implemented;
- Timing in approaching and engaging the community is crucial: Project executives should inform communities from an early stage about the temporal dimensions of planning and implementing ancillary infrastructure, government authorities should ensure due process and opportunities for appeal by those affected;
- Ancillary infrastructure should be planned in a participatory manner in order to be adapted to local needs;
- Project executives should not approach communities as homogenous entities: power differentials within communities, such as across age or gender differences, as well as potential problems of elite capture need to be taken into account;
- Project executives need to take the concerns and (historical) grievances of affected communities seriously;
- Future local appropriation and modification of ancillary infrastructure should be taken into account by project planners;
- Public authorities must ensure that ancillary infrastructure is included in impact assessments (e.g. ESIAs) and that it and its potential impacts are integrated into project presentations;
- Executing companies must ensure proper cultural sensitivity training for (foreign/non-local) staff deployed to work on the construction, as well as establish clear protocols for dealing with conflicts, if they arise;
- In general: ancillary infrastructure should be conceived as a modular component of LSREPs, i.e., as self-sustained infrastructure with its own life cycle, which has social and ecological impacts, even if the primary infrastructure project does not materialize.

CRediT authorship contribution statement

Clemens Greiner: Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Conceptualization. **Britta Klagge:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgements

We thank our interlocutors for sharing their knowledge and experiences, three anonymous reviewers and the editor of this journal for constructive comments. We also thank Jonathan DeVore for proof reading. This research and publication was faciltiated by the German Research Foundation (DFG) through funding for the project "Energy Futures" as part of the Collaborative Research Center "Future Rural Africa" (Project-ID 328966760; TRR 228/2).

References

Abram, Simone, 2011. Culture and Planning. Ashgate Pub. Company.
Abram, Simone, 2014. The time it takes: temporalities of planning. J. Roy. Anthropol. Inst. 20 (S1), 129–147. https://doi.org/10.1111/1467-9655.12097.
Abram, Simone, Weszkalnys, Gisa, 2011. Introduction: anthropologies of planning—temporality, imagination, and ethnography. Focaal 2011 (61), 3–18. https://doi.org/10.3167/fcl.2011.610101.
Abram, Simone, Weszkalnys, Gisa (Eds.), 2013. Elusive Promises: Planning in the Contemporary World. Berghahn Books.
Ahmed, Abubakari, Gasparatos, Alexandros, 2020. Multi-dimensional energy poverty patterns around industrial crop projects in Ghana: enhancing the energy poverty alleviation potential of rural development strategies. Energy Pol. 137 https://doi.org/10.1016/j.enpol.2019.111123.
Alamgir, M., Campbell, M.J., Sloan, S., Goosem, M., Clements, G.R., Mahmoud, M.I., Laurance, W.F., 2017. Economic, socio-political and environmental risks of road development in the tropics. Curr. Biol. 27 (20), R1130–R1140. https://doi.org/10.1016/j.cub.2017.08.067.
Ansar, Atif, Flyvbjerg, Bent, Budzier, Alexander, Lunn, Daniel, 2016. Does infrastructure investment lead to economic growth or economic fragility? Evidence from China. Oxf. Rev. Econ. Pol. 32 (3), 360–390. https://doi.org/10.1093/oxrep/grw022.
Aung, Thiri Shwesin, Fischer, Thomas B., Azmi, Azlin Suhaida, 2021. Social impacts of large-scale hydropower project in Myanmar: a social life cycle assessment of Shweli hydropower dam 1. Int. J. Life Cycle Assess. 26 (2), 417–433. https://doi.org/10.1007/s11367-021-01868-3.
Bakken, Tor Haakon, Sundt, Håkon, Ruud, Audun, Harby, Atle, 2012. Development of small versus large hydropower in Norway– comparison of environmental impacts. Energy Proc. 20, 185–199. https://doi.org/10.1016/j.egypro.2012.03.019.
Bear, Laura, 2016. Time as technique. Annu. Rev. Anthropol. 45 (1), 487–502. https://doi.org/10.1146/annurev-anthro-102313-030159.
Becker, Dennis R., Harris, Charles C., McLaughlin, William J., Nielsen, Erik A., 2003. A participatory approach to social impact assessment: the interactive community forum. Environ. Impact Assess. Rev. 23 (3), 367–382. https://doi.org/10.1016/s0195-9255(02)00098-7.
Bollig, Michael, Greiner, Clemens, Österle, Matthias, 2014. Inscribing identity and agency on the landscape: of pathways, places, and the transition of the public sphere in East Pokot, Kenya. Afr. Stud. Rev. 57 (3), 55–78. https://doi.org/10.1017/asr.2014.92.
Boyer, Dominic, 2019. Energopolitics. Wind and Power in the Anthropocene. Duke University Press, Durham.
Brookes, Naomi, Sage, Daniel, Dainty, Andrew, Locatelli, Giorgio, Whyte, Jennifer, 2017. An island of constancy in a sea of change: rethinking project temporalities with long-term megaprojects. Int. J. Proj. Manag. 35 (7), 1213–1224. https://doi.org/10.1016/j.ijproman.2017.05.007.
Brunet, Carole, Savadogo, Oumarou, Baptiste, Pierre, Bouchard, Michel A., Rakotoary, Jean Chrysostome, Ravoninjatovo, Andry, Cholez, Céline, Gendron, Corinne, Merveille, Nicolas, 2020. Impacts generated by a large-scale solar photovoltaic power plant can lead to conflicts between sustainable development goals: a review of key lessons learned in Madagascar. Sustainability 12 (18). https://doi.org/10.3390/su12187471.
Callaghan, George, Williams, Derek, 2014. Teddy bears and tigers: how renewable energy can revitalise local communities. Local Econ. J. Local Econ. Policy U. 29 (6–7), 657–674. https://doi.org/10.1177/0269094214551254.
Catolico, A.C.C., Maestrini, M., Strauch, J.C.M., Giusti, F., Hunt, J., 2021. Socioeconomic impacts of large hydroelectric power plants in Brazil: a synthetic control assessment of Estreito hydropower plant. Renew. Sustain. Energy Rev. 151 https://doi.org/10.1016/j.rser.2021.111508.
Corry, Robert C., 2011. A case study on visual impact assessment for wind energy development. Impact Assess. Proj. Apprais. 29 (4), 303–315. https://doi.org/10.3152/146155111X12959673796047.
Crootof, Arica, Shrestha, Rashmi, Albrecht, Tamee, Ptak, Thomas, Scott, Christopher A., 2021. Sacrificing the local to support the national: politics, sustainability, and governance in Nepal's hydropower paradox. Energy Res. Social Sci. 80 https://doi.org/10.1016/j.erss.2021.102206.
da Silva, Pimentel Diogo Gardenio, Parkins, John R., Sherren, Kate, 2021. Do methods used in social impact assessment adequately capture impacts? An exploration of the research-practice gap using hydroelectricity in Canada. Energy Res. Social Sci. 79 https://doi.org/10.1016/j.erss.2021.102188.
Danwatch. A people in the way of progress. Prostitution, alcoholism and a lawsuit on illegal land acquisition in the Lake Turkana Wind Power project. 1–40. https://old.danwatch.dk/wp-content/uploads/2016/05/Danwatch_report_A-PEOPLE-IN-THE-WAY-OF-PROGRESS-2016_web.pdf.
del Río, P., Burguillo, M., 2008. Assessing the impact of renewable energy deployment on local sustainability: towards a theoretical framework. Renew. Sustain. Energy Rev. 12 (5), 1325–1344. https://doi.org/10.1016/j.rser.2007.03.004.
Delina, Laurence L., Sovacool, Benjamin K., 2018. Of temporality and plurality: an epistemic and governance agenda for accelerating just transitions for energy access and sustainable development. Curr. Opin. Environ. Sustain. 34, 1–6. https://doi.org/10.1016/j.cosust.2018.05.016.
Dunlap, Alexander, 2021. Spreading 'green' infrastructural harm: mapping conflicts and socio-ecological disruptions within the European Union's transnational energy grid. Globalizations 1–25. https://doi.org/10.1080/14747731.2021.1996518.
Engen, Sigrid, Vera, Helene Hausner, Fauchald, Per, Ruud, Audun, Broderstad, Else Grete, 2023. Small hydropower, large obstacle? Exploring land use conflict, Indigenous opposition and acceptance in the Norwegian Arctic. Energy Res. Social Sci. 95 https://doi.org/10.1016/j.erss.2022.102888.
Flyvbjerg, Bent, 2014. What you should know about megaprojects and why: an overview. Proj. Manag. J. 45 (2), 6–19. https://doi.org/10.1002/pmj.21409.
Gasparatos, A., Ahmed, A., Voigt, C., 2021. Facilitating policy responses for renewable energy and biodiversity. Trends Ecol. Evol. https://doi.org/10.1016/j.tree.2021.01.013.
Góes, Maria de Fátima Barbosa, Andrade, José Célio Silveira, Jabbour, Charbel Jose Chiappetta, Silva, Marcelo Santana, 2021. Wind power projects in Brazil: challenges and opportunities increasing co-benefits and implications for climate and energy policies. Environ. Dev. Sustain. 23 (10), 15341–15367. https://doi.org/10.1007/s10668-021-01300-8.
Greiner, C., 2013. Guns, land and votes: cattle rustling and the politics of boundary-(re)making in Northern Kenya. Afr. Affairs 112 (447), 216–237.
Greiner, Clemens, 2020. Negotiating access to land & resources at the geothermal frontier in Baringo, Kenya. In: Lind, Jeremy, Okenwa, Doris, Scoones, Ian (Eds.), Land, Investment & Politics: Reconfiguring Eastern Africa's Pastoral Drylands. James Currey, Woodbridge, pp. 101–109.
Greiner, Clemens, Greven, David, Klagge, Britta, 2021a. Roads to change: livelihoods, land disputes, and anticipation of future developments in rural Kenya. Eur. J. Dev. Res. 33 (4), 1044–1068. https://doi.org/10.1057/s41287-021-00396-y.
Greiner, Clemens, Vehrs, Hauke, Bollig, Michael, 2021b. Land-use and land-cover changes in pastoral drylands: long-term dynamics, economic change, and shifting socioecological frontiers in Baringo, Kenya. Hum. Ecol. 49 (5), 565–577. https://doi.org/10.1007/s10745-021-00263-8.
Greiner, Clemens, Klagge, Britta, Owino, Evelyne Atieno, 2023. The political ecology of geothermal development: green sacrifice zones or energy landscapes of value? Energy Res. Social Sci. 99, 103063 https://doi.org/10.1016/j.erss.2023.103063.
Greven, David, 2023. Bursting pipes and broken dreams: on ruination and reappropriation of large-scale water infrastructure in Baringo County, Kenya. Journal of Eastern African Studies 1–21. https://doi.org/10.1080/17531055.2023.2231790.
Grubert, E., Zacarias, M., 2022. Paradigm shifts for environmental assessment of decarbonizing energy systems: emerging dominance of embodied impacts and design-oriented decision support needs. Renew. Sustain. Energy Rev. 159, 112208 https://doi.org/10.1016/j.rser.2022.112208.
Guerin, Turlough F., 2019. Impacts and opportunities from large-scale solar photovoltaic (PV) electricity generation on agricultural production. Environ. Qual. Manag. https://doi.org/10.1002/tqem.21629.
Hetherington, Kregg, 2017. Surveying the future perfect. Infrastructures and Social Complexity 40–50. https://doi.org/10.9774/GLEAF.9781315622880_4.
Howe, C., Lockrem, J., Appel, H., Hackett, E., Boyer, Dominic, Hall, R., Schneider-Mayerson, M., Pope, A., Gupta, A., Rodwell, E., Ballestero, A., Durbin, T., el-Dahdah, F., Long, E., Mody, C., 2015. Paradoxical infrastructures: ruins, retrofit, and risk. Sci. Technol. Hum. Val. https://doi.org/10.1177/0162243915620017.
Kelly, Sarah, 2019. Megawatts mask impacts: small hydropower and knowledge politics in the Puelwillimapu, Southern Chile. Energy Res. Social Sci. 54, 224–235. https://doi.org/10.1016/j.erss.2019.04.014.
Kelly-Richards, Sarah, Silber-Coats, Noah, Crootof, Arica, Tecklin, David, Bauer, Carl, 2017. Governing the transition to renewable energy: a review of impacts and policy issues in the small hydropower boom. Energy Pol. 101, 251–264. https://doi.org/10.1016/j.enpol.2016.11.035.

Klagge, Britta, 2021. The renewable energy revolution: risk, investor and financing structures—with case studies from Germany and Kenya. In: Knox-Hayes, Janelle, Wójcik, Dariusz (Eds.), The Routledge Handbook of Financial Geography. Routledge, New York, pp. 620–645.

Klagge, Britta, Nweke-Eze, Chigozie, 2020. Financing large-scale renewable-energy projects in Kenya: investor types, international connections, and financialization. Geogr. Ann. B Hum. Geogr. 1–23. https://doi.org/10.1080/04353684.2020.1729662.

Klagge, Britta, Greiner, Clemens, Greven, David, Nweke-Eze, Chigozie, 2020. Cross-scale linkages of centralized electricity generation: geothermal development and investor–community relations in Kenya. Polit. Govern. 8 (3), 211–222. https://doi.org/10.17645/pag.v8i3.2981.

Knuth, Sarah, Behrsin, Ingrid, Levenda, Anthony, McCarthy, James, 2022. New political ecologies of renewable energy. Environ. Plann.: Nature and Space 5 (3), 997–1013. https://doi.org/10.1177/25148486221108164.

Kochore, Hassan H., 2016. The road to Kenya? Visions, expectations and anxieties around new infrastructure development in Northern Kenya. Journal of Eastern African Studies 10 (3), 494–510. https://doi.org/10.1080/17531055.2016.1266198.

Kramarz, Teresa, Park, Susan, Craig, Johnson, 2021. Governing the dark side of renewable energy: a typology of global displacements. Energy Res. Social Sci. 74 https://doi.org/10.1016/j.erss.2020.101902.

Küpers, Sophia, Batel, Susana, 2023. Time, history and meaning-making in research on people's relations with renewable energy technologies (RETs) – a conceptual proposal. Energy Pol. 173 https://doi.org/10.1016/j.enpol.2022.113358.

Labussière, Olivier, Nadaï, Alain, 2018. The temporalities of energy transition processes. In: Labussière, Olivier, Nadaï, Alain (Eds.), Energy Transitions. A Socio-Technical Inquiry. Palgrave Macmillan, Cham, pp. 277–317.

Larkin, Brian, 2013. The politics and poetics of infrastructure. Annu. Rev. Anthropol. 42 (1), 327–343. https://doi.org/10.1146/annurev-anthro-092412-155522.

Laurance, W.F., Peletier-Jellema, A., Geenen, B., Koster, H., Verweij, P., Van Dijck, P., Lovejoy, T.E., Schleicher, J., Van Kuijk, M., 2015. Reducing the global environmental impacts of rapid infrastructure expansion. Curr. Biol. 25 (7), R259–R262. https://doi.org/10.1016/j.cub.2015.02.050.

Laurian, Lucie, Inch, Andy, 2018. On time and planning: opening futures by cultivating a "sense of now". J. Plann. Lit. 34 (3), 267–285. https://doi.org/10.1177/0885412218817775.

Lawhon, Mary, Follmann, Alexander, Braun, Boris, Cornea, Natasha, Greiner, Clemens, Guma, Prince, Karpouzoglou, Timos, Revilla Diez, Javier, Schindler, Seth, Schramm, Sophie, Sielker, Franziska, Tups, Gideon, Vij, Sumit, Dannenberg, Peter, 2023. Making heterogeneous infrastructure futures in and beyond the Global South. Futures, 103270. https://doi.org/10.1016/j.futures.2023.103270.

Lema, Rasmus, Lakshmi Bhamidipati, Padmasai, Gregersen, Cecilia, Ulrich, Elmer Hansen, Kirchherr, Julian, 2021. China's investments in renewable energy in Africa: creating co-benefits or just cashing-in? World Dev. 141 https://doi.org/10.1016/j.worlddev.2020.105365.

McCarthy, James, 2015. A socioecological fix to capitalist crisis and climate change? The possibilities and limits of renewable energy. Environ. Plann.: Econ. Space 47 (12), 2485–2502. https://doi.org/10.1177/0308518x15602491.

O'Sullivan, Kate, Golubchikov, Oleg, Mehmood, Abid, 2020. Uneven energy transitions: understanding continued energy peripheralization in rural communities. Energy Pol. 138 https://doi.org/10.1016/j.enpol.2020.111288.

Pasqualetti, Martin, 2011. Social barriers to renewable energy landscapes. Geogr. Rev. 101 (2), 201–223. https://doi.org/10.1111/j.1931-0846.2011.00087.x.

Pearse, Rebecca, Bryant, Gareth, 2022. Labour in transition: a value-theoretical approach to renewable energy labour. Environ. Plann.: Nature and Space 5 (4), 1872–1894. https://doi.org/10.1177/25148486211055542.

Ramakrishnan, Kavita, O' Reilly, Kathleen, Budds, Jessica, 2020. The temporal fragility of infrastructure: theorizing decay, maintenance, and repair. Environ. Plann.: Nature and Space. https://doi.org/10.1177/2514848620979712.

Rodríguez-Segura, Francisco Javier, Juan Carlos, Osorio-Aravena, Frolova, Marina, Terrados-Cepeda, Julio, Muñoz-Cerón, Emilio, 2023. Social acceptance of renewable energy development in southern Spain: exploring tendencies, locations, criteria and situations. Energy Pol. 173, 113356 https://doi.org/10.1016/j.enpol.2022.113356.

Rudman, Justine, Gauché, Paul, Esler, Karen J., 2017. Direct environmental impacts of solar power in two arid biomes: an initial investigation. South Afr. J. Sci. 113 (11/12) https://doi.org/10.17159/sajs.2017/20170113.

Rudolph, David, Clausen, Laura Tolnov, 2021. Getting used to it, but …? Rethinking the elusive U-curve of acceptance and post-construction assumptions. In: A Critical Approach to the Social Acceptance of Renewable Energy Infrastructures, pp. 63–81.

Schulz, Christopher, Saklani, Udisha, 2021. The future of hydropower development in Nepal: views from the private sector. Renew. Energy 179, 1578–1588. https://doi.org/10.1016/j.renene.2021.07.138.

Sovacool, Benjamin K., Bulan, L.C., 2012. Energy security and hydropower development in Malaysia: the drivers and challenges facing the Sarawak Corridor of Renewable Energy (SCORE). Renew. Energy 40 (1), 113–129. https://doi.org/10.1016/j.renene.2011.09.032.

Tang, Amy, Taylor, John E., Mahalingam, Ashwin, 2013. Strategic structure matrix: a framework for explaining the impact of superstructure organizations on the diffusion of wind energy infrastructure. Energy Pol. 63, 69–80. https://doi.org/10.1016/j.enpol.2013.08.029.

Terrapon-Pfaff, Julia, Fink, Thomas, Viebahn, Peter, Jamea, El Mostafa, 2019. Social impacts of large-scale solar thermal power plants: assessment results for the NOORO I power plant in Morocco. Renew. Sustain. Energy Rev. 113, 109259 https://doi.org/10.1016/j.rser.2019.109259.

van de Ven, D.J., Capellan-Perez, I., Arto, I., Cazcarro, I., de Castro, C., Patel, P., Gonzalez-Eguino, M., 2021. The potential land requirements and related land use change emissions of solar energy. Sci. Rep. 11 (1), 2907. https://doi.org/10.1038/s41598-021-82042-5.

Vehrs, Hauke-Peter, 2022. Pokot Pastoralism: Environmental Change and Socio-Economic Transformation in North-west Kenya. James Currey.

Wilson, Geoff A., Dyke, Sarah L., 2016. Pre- and post-installation community perceptions of wind farm projects: the case of Roskrow Barton (Cornwall, UK). Land Use Pol. 52, 287–296. https://doi.org/10.1016/j.landusepol.2015.12.008.

Wolsink, Maarten, 2007. Planning of renewables schemes: deliberative and fair decision-making on landscape issues instead of reproachful accusations of non-cooperation. Energy Pol. 35 (5), 2692–2704. https://doi.org/10.1016/j.enpol.2006.12.002.